Per-site ΔRMSE from adding mortality — C+T vs C+T+M

Below 0 = mortality reduces the site's RMSE | bars coloured by relative mortality class (site maximum) | repeated CV (3×80 %) | tree cover ≥30 % (104 sites, 381 site-years) · deadtrees v4.1

ΔRMSE = RMSE(with M) − RMSE(without M)

	XGBoost (Optuna) · 12m	XGBoost (Optuna) · 24m	LightGBM (Optuna) · 12m	LightGBM (Optuna) · 24m
GPPsat (µmol m⁻² s⁻¹)	62% improved	60% improved	64% improved	62% improved
NEPmax (µmol m⁻² s⁻¹)	74% improved	73% improved	73% improved	70% improved
ETmax (mm d⁻¹)	57% improved	56% improved	59% improved	53% improved
WUE (g C mm⁻¹)	54% improved	67% improved	57% improved	55% improved

Relative mortality
site maximum, mean of 100-500 m buffers
(breaks: mean ± 0.5 SD)

- Low (< 9.4 %, 41 sites)
- Mid (9.4–20.6 %, 40 sites)
- High (> 20.6 %, 23 sites)

Sites (sorted by ΔRMSE within each panel)